AI4RO1 Group Assignment Description

Each group should consider a domain mentioned in the following and provide an ontology that classifies all the entities and relationships that are potentially relevant for a robot accomplishing a given task in the domain. Each group is assigned a domain according to the rule **group *n* – scenario *n***. The choice of scenarios is fixed, i.e., it is not possible for group ***n*** to work on a scenario ***m*** with ***n≠m***.

For each domain you should come up with a choice of entities and relationship that is complete enough for a robot to "understand" all the entities it may come in contact with, the relationships that might help the robot to accomplish the task and the data properties of entities that you deem necessary. You are free to search on the web for documentation related to the specific domain and also refer to off-the-shelf ontologies when relevant.

The hand-in for the assignment is the ontology itself (in OWL/XML format) and a small report of 10 pages at most. In the report you should clearly indicate the rationale behing your design choices and acknowledge all external sources, e.g., documentation, existing ontologies and such. The usage of chatbots to complete the assignment is not forbidden, but it is warmly discouraged: it is much better to hand in a small and simple ontology that you can understand and motivate, rather than a big and fancy ontology of which you now little or nothing.

AI4RO1 Group Assignment Domains

1. *Automated assembly lines for electronics*: the task is to assemble a circuit board given a set of discrete components (chips, transistor, resistors, etc.) and some placement rules.
2. *Welding robots in automotive plants*: the task is to weld together parts of a car frame given a set of parts and some placement rules. The fact that some parts need to be soldered before other should be taken into account.
3. *Packaging and palletizing robots*: the task is to assemble pallets of boxes of predefined size. Consider difference in size and weight of boxes, e.g., heavy and large boxes should never be places on top of small and light ones.

4. *Hospital delivery robots for medicines and supplies*: the task is to deliver medications to patients in their rooms located across multiple floors. Different kinds of medications should be considered (e.g., analgesic, blood thinners, laxative, etc.) and properties to define which patients take which prescriptions.

5. *Automated Guided Vehicles (AGVs) for material transport in mining sites*: the task is to bring ore from the excavation site to the metal-extraction site and to transport residual waste to the disposal site for recycling. The excavation site is comprised of multiple tunnels situated on several levels. Maximum payload and other physical elements should be taken into account.

6. *Robotic arms for order picking*: the task is to select the right parcel to deliver from stored goods. Different goods need to be picked with different end-effectors which are mounted on different arms. Cooperation between arms could be required for heavy items.

7. *Drones for inventory scanning:* the task is to have drones travel in a warehouse were goods are stored. Each item is identified by a unique OCR code (e.g., a QR code) that the drone can read to assess the consistency of the inventory. Goods are stored on shelves placed in rows. Drones fly among rows and home at a recharging station where they can also transmit data.

8. *Autonomous tractors for plowing, planting and irrigating*: the task is to prepare the terrain, plant various crops and irrigate the offspring vegetables. Different vehicles perform different operations. A taxonomy of crops and their features is required, as well as the description of fields and the kind of crops planted and the way they should be irrigated.

9. *Bricklaying robots for building walls*: the task is to build walls of specific size considering various kinds of bricks (burnt clay, concrete, sand lime, plaster, etc.) and various kind of mortar (cement, lime, etc.).

10. *Inspection drones for bridges*: the task is to inspect various parts of the building and perform specific checks depending on the part (e.g., bolt, cable, beam, etc.). Properties should include how various elements are connected and what peripheral can be used to perform a specific check.

11. *Demolition robots for controlled dismantling*: the task is to take apart a building by removing its various parts in a controlled fashion. Different robots perform different removal tasks (e.g. windows, electrical cables, plumbing, inner walls, tiles, etc.) and the topology of the building should be specified.

12. *Firefighting robots for hazardous environments*: the task is to extinguish fire in the presence of hazardous equipment, harmful chemicals or radiations. Different fire suppression strategies should be considered (e.g., carbon dioxide, powder, explosives, etc.) in presence of different environments (e.g., electrical equipment, radioactive material, solvents, etc.)
13. Surveillance robots for building security: the task is to patrol various rooms in a building as well as external premises. The different kinds of rooms should be specified (e.g., office, warehouse, production plant, etc.) as well as the relationship among different rooms in the building.
14. Customer assistance robots in stores: the task is to show customers various goods and help them make a selection describing various features. Choose a specific application (e.g., convenience store, DIY store, sporting apparel) and consider a taxonomy of goods and potential relationships between them (e.g., beer, pizza and chips for a party, nuts and bolts of matching size, gear for running, biking, etc.)
15. Autonomous delivery robots for food and parcels: the task is to deliver a parcel at a specific address. Buildings and streets must be described as well as their relationships. Different kinds of goods and customers should be taken into account.
16. Robotic vacuum cleaners: the task is to clean a house with various rooms inside which there are pieces of furniture. Modeling of specific circumstances (e.g., wet floor, carpets, stairwells) should be considered as well as robot-specific items like the docking station.
17. Cooking robots for automated meal preparation: the task is to prepare simple dishes starting from basic ingredients. Various preparations (e.g., vanilla cake, roastbeef, eggplant parmesan) should be considered together with the ingredients and relationships among them.